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PI Name:	Turner, Russell T Ph.D.		
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PI Email:	Russell.Turner@oregonstate.edu	**Fax:**	FY 541 737 6914
PI Organization Type:	UNIVERSITY	**Phone:**	541 737 9545
Organization Name:	Oregon State University		
PI Address 1:	Nutrition & Exercise Sciences/Skeletal Biology Laboratory		
PI Address 2:	Milam Hall, Mail Stop 107B		
PI Web Page:			
City:	Corvallis	**State:**	OR
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	We have previously obtained evidence suggesting that mechanical loading acts on bone cells via a cascade mechanism. We have postulated that the initial step of the cascade initiated by dynamic loading involves mechanoreceptor-transduced regulation of expression of a small number of immediate response genes which code for transcription factors and other cell cycle regulatory factors. These factors regulate expression of genes which code for skeletal signaling molecules (e.g., growth factors) which in turn influence the expression of genes to control cell proliferation, metabolism and expression of differentiated bone cell function. If this model is correct, disuse osteopenia is largely mediated by altered expression of cell signaling molecules and may be corrected by pharmacological agents which modulate the skeletal expression of these intermediates. Studies performed during the previous funding interval have revealed that the signal transduction pathways for parathyroid hormone (PTH) and estrogen partially overlap with		

Task Description:	weight bearing in that each of these agents regulates the expression of skeletal signaling molecules. These studies also demonstrated an unexpected pronounced gender difference in the skeletal response to decreased mechanical usage. The proposed research is designed to test the hypothesis that it is possible to bypass the mechanoreceptor step in hindlimb unloaded rats by pharmacological intervention with PTH and a tissue-selective estrogen agonist (clomiphene) with little activity on reproductive tissues. The specific aims of the proposed studies in hindlimb elevated rats are to determine: 1) the comparative effects of skeletal unloading on normal and gonadectomized males and females; 2) the efficacy of clomiphene treatment to prevent cancellous osteopenia and maintain normal bone turnover, architecture and mechanical properties; and 3) if the skeletal efficacy of combined treatment with clomiphene and human PTH (hPTH) is superior to monotherapy.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although these studies were directed at developing a method to prevent bone loss during spaceflight, the results are highly relevant to human disease. In the course of these studies, we identified PDGF-A as a putative causative factor for parathyroid hormone bone disease and established that antagonism of PDGF-A signaling can greatly reduce hypercalcemia, focal bone resorption, and osteitis fibrosa in an animal model for hyperparathyroidism. If confirmed in humans, drugs that target PDGF-A signaling could be used to treated patients with renal osteodystrophy as well as individuals with primary and other forms of secondary hyperparathyroidism. We have also shown that "bone specific" and "kidney specific" genes are expressed in white blood cells and respond in an analogous manner to PTH as in bone and kidney. These findings suggest that white blood cells can be used as a surrogate tissue to monitor positive as well as adverse responses to systemic factors in organs which cannot be directly evaluated without use of invasive procedures.
	Martin, et al: Bone 32:261-267, 2003: Although the rat is the most common animal model used in studying osteoporosis, it is often used inappropriately. Osteoporosis is a disease that most commonly occurs in humans long after growth plate fusion with the associated cessation of longitudinal bone growth, but there has been a question as to when or to what extent the rat growth plate fuses. To investigate this question, we used microcomputed X-ray tomography, at voxel resolutions ranging from (5.7 µm)3 to (11 µm)3, to image the proximal epiphyseal growth plates of both male (n=19) and female (n=15) rat tibiae, ranging in age from 2 to 25 months. The three-dimensional images were used to evaluate fusion of the epiphyseal growth plate by quantitating the amount of cancellous bone that has bridged across the growth plate. The results suggest that the time course of fusion of the epiphyseal growth plate follows a sigmoidal pattern and 90% of the bridges form 6.5 months in females and 7.4 months in males. Lotinun and Turner: Endocrinology 144:2000-2007, 2003: Parathyroid bone disease in humans is caused by chronic hyperparathyroidism (HPT). Continuous infusion of PTH into rats results in histological changes similar to parathyroid bone disease, including increased bone formation, focal bone resorption, and severe peritrabecular fibrosis, whereas pulsatile PTH increases bone formation without skeletal abnormalities. Using a cDNA microarray with over 5000 genes, we identified an association between increased platelet-derived growth factor-A (PDGF-A) signaling and PTH-induced bone disease in rats. Verification of PDGF-A overexpression was accomplished with a ribonuclease protection assay. Using immunohistochemistry, PDGF-A peptide was localized to mast cells in PTH-treated rats. We also report a novel strategy for prevention of parathyroid bone disease using triazolopyrimidine (trapidil). Trapidil, an inhibitor of PDGF signaling, did not have any effect on indexes of bone turnover in normal rats. However, dramatic reductions in marrow fibrosis and bone resorption, but not bone formation, were observed in PTH-treated rats given trapidil. Also, trapidil antagonized the PTH-induced increases in mRNA levels for PDGF-A. These results suggest that PDGF signaling is important for the detrimental skeletal effects of HPT, and the drugs that target the cytokine or its receptor might be useful in reducing or preventing parathyroid bone disease. Lotinun, et al: Bone 33:124-131, 2003: 16a-Hydroxyestrone (16a-OHE1), an endogenous estrogen metabolite, is associated with increased bone density in postmenopausal women. This study was designed to evaluate the long-term activity of this metabolite on bone, uterus, and serum cholesterol in an animal model for postmenopausal bone loss. A preliminary dose-response study performed in weanling rats determined 2000 µg/kg/day to be the optimal dose of 16a-OHE1 for studying estrogenic effect on bone. The long-term experiment was performed in 6-month-old animals that were either sham-operated or OVX. The OVX rats were implanted sc with 60-day continuous-release carrier, 17ß-estradiol (E2) (33 µg/kg/day) or 16a-OHE1 pellets (2000 µg/kg/day). OVX decreased uterine weight, increased body weight, serum cholesterol, and all dynamic bone histomorphometric measurements in cortical and cancellous bone, and resulted in a 54% bone loss at the tibial metaphysis. E2 completely prevented OVX-induced bone loss, suppressed bone turnover, and induced uterine hypertrophy and hypercholesterolemia. 16a-OHE1 acted as an E2 agonist on bone, suppressing bone formation and resorption. However, the estrogen metabolite lowered serum cholesterol and was only a partial E2 agonist on uterine weight and epithelial cell height. These results suggest that 16a-OHE1 is an estrogen agonist on bone and may be responsible, in part, for the cholesterol-lowering activity attributed to estrogen. As a consequence of its skeletal effects, older women who produce high levels of 16a-OHE1 could have a lower risk for developing postmenopausal osteoporosis than women who produce less-active estrogen metabolites. Maran et al; J Musculoskel Neuron Interact 3:83-88, 2003: An important consideration in interpreting indices of gene expression in human bone is relating mRNA levels to functional endpoints such as bone architecture. In the present study, a method was developed for quantitative measurement of gene expression and bone morphology in the same specimen. Three-dimensional images of iliac crest bone biopsies from healthy premenopausal women were obtained using a novel high resolution cryogenic µ-CT scanner. RNA was isolated from the biopsies and mRNA levels were measured for genes related to bone metabolism. The gene expression profile and variability of expression within iliac crest biopsies of women was similar to human osteoblastic cell lines and rat long bones. mRNA for alkaline phosphatase, bone matrix proteins, and selected cytokines and cytokine receptors were consistently detected in biopsies. As previously shown in rat bone, there was a tight correlation between mRNA levels for type 1 collagen and osteocalcin and no correlation between bone matrix proteins and alkaline phosphatase. The relative abundance of the mRNA for the three most prevalent transforming growth factor-ß (TGF-ß) isoforms in bone (TGF-ß1 >> TGF-ß3 > TGF-ß2) was the same as the known abundance of the corresponding TGF-ß peptides in bone matrix. The results demonstrate the feasibility of analyzing the three-dimensional architecture of a bone biopsy using cryogenic µ-CT imaging and then measuring expression of genes related to bone cell function within the same specimen following RNA extraction and analysis. Hefferan, et al: Alcohol Clin Exp Res 27:111-117, 2003: Background: Alcohol abuse is associated with an increased risk

Task Progress:	for osteoporosis. However, co-morbidity factors may play an important role in the pathogenesis of alcohol-related bone fractures. Sub-optimal mechanical loading of the skeleton, an established risk factor for bone loss, may occur in some alcohol abusers due to reduced physical activity, muscle atrophy or both. The effect of alcohol consumption and reduced physical activity on bone metabolism has not been well studied. The purpose of the present study was to determine if mechanical disuse alters bone metabolism in a rat model for chronic alcohol abuse. Alcohol was administered in the diet (35% caloric intake) of 6-month-old male rats for 4 weeks. Rats were hindlimb unloaded the final 2 weeks of the experiment in order to prevent dynamic weight bearing. Afterwards, cortical bone histomorphometry was evaluated at the tibia-fibula synostosis. At the periosteal surface of the tibial diaphysis, alcohol and hindlimb unloading independently decreased mineralizing perimeter, mineral apposition rate and bone formation rate. In addition, alcohol, but not hindlimb unloading, increased endocortical bone resorption. The respective detrimental effects of alcohol and hindlimb unloading to inhibit bone formation were additive; there was no interaction between the two variables. Reduced weight bearing accentuates the detrimental effects of alcohol on cortical bone in adult male rats by further inhibiting bone formation. This finding suggests that reduced physical activity may be a co-morbidity factor for osteoporosis in alcohol abusers. Hefferan, et al: J Appl Physiol, 95:1775-1780, 2003: Prolonged spaceflight results in bone loss in astronauts but there is considerable individual variation. The goal of this rat study was to determine whether gender influences bone loss during simulated weightlessness. Six-month-old Fisher 344 rats were hindlimb unweighted for 2 weeks, after which the proximal tibiae were evaluated by histomorphometry. There were gender differences in tibia length, bone area, cancellous bone architecture, and bone formation. Compared to the females, the males had a 11.6% longer tibiae, a 27.8% greater cortical bone area, and a 37.6% greater trabecular separation. Conversely, the females had greater cortical (316%) and cancellous (145%) bone formation rates, 28.6% more cancellous bone, and 30% greater trabecular number. Hindlimb unweighting resulted in large reductions in periosteal bone formation and mineral apposition rate in both genders. Unweighting also caused cancellous bone loss in both genders; trabecular number was decreased and trabecular separation was increased. There was, however, no change in trabecular thickness in either gender. These architectural changes in cancellous bone were associated with decreases in bone formation and steady-state mRNA levels for bone matrix proteins, and cancellous bone resorption. In conclusion, there are major gender-related differences in bone mass and turnover; however, the bone loss in hindlimb unweighted adult male and female rats appears to be due to similar mechanisms. Maran, et al: Calcif Tissue Intl, 74:204-207, 2004.: Glyceraldehyde 3-phosphate dehydrogenase (GAPDH) mRNA levels are commonly used as an internal control to normalize gene expression data based on the belief that this gene is constitutively expressed. However, GAPDH mRNA levels increased by more than 2.5 fold in tibiae of hind-limb unloaded female rats compared to L32 mRNA levels as determined by 2 independent (northern analysis and ribonuclease protection assay) methods. Similarly, GAPDH mRNA levels increased compared to 18S ribosomal RNA. Normal aging, ovariectomy, and treatment with estrogens, growth hormone and alcohol do not show any disparity in GAPDH mRNA levels, whereas, in some experiments parathyroid hormone increased GAPDH mRNA levels. Taken together, these findings indicate that it is essential to demonstrate that GAPDH expression is not altered prior to using the gene as an internal control. Our results also warrant further investigation of the role of GAPDH in mechanical signaling. Sibonga et al., Endocrinology 144:785-792, 2003.: Estrogens are potent inhibitors of bone resorption, however, the detrimental effects of hormone replacement therapy in postmenopausal osteoporotic women exceeds the benefits. In three experiments, we evaluated the pharmacological effects of 2-methoxyestradiol (2ME2), a natural metabolite of estradiol with antitumor activity, on several estrogen target tissues. Experiment 1: we gavaged recently ovariectomized (OVX'd) 9.5-week-old rats with 2ME2 at doses of 0, 0.1, 1, 4, 20, and 75 mg/kg in a 21-day dose-response study. 2ME2 reduced body weight and serum cholesterol, increased uterine weight and epithelial cell height, and inhibited longitudinal and radial bone growth compared to values in the untreated OVX'd rats. All doses of 2ME2 maintained cancellous bone mass at the baseline level, the lowest effective dose being 20-fold less than an uterotrophic dose. Experiment 2: in an 8-week experiment in adult OVX'd rats, a non-uterotrophic dose of 2ME2 (4 mg/kg/d) suppressed body weight gain, inhibited bone formation in cancellous bone and partially prevented bone loss in the tibial metaphysis. Experiment 3: in weanling rats ICI 182,780 [ICI] did not antagonize the effect of 2ME2. We conclude that 2ME2 antagonizes the skeletal changes that follow OVX at doses that have minimal or no effects in the uterus in both young and adult rats; 2ME2 does not appear to act via estrogen receptors and is active on bone at doses well below those required for tumor suppression in mice. 2ME2 through a novel pathway, may be a useful alternative to conventional hormone replacement therapy for prevention of postmenopausal bone loss. In addition, 2ME2 may be useful as a chemopreventive agent for spaceflight-induced bone loss due to weightlessness and cancer due to radiation exposure. Lotinun et al., J Bone Miner Res 19:1165-1171, 2004: PTH increases cancellous bone volume but also increases cortical bone turnover which could have negative effects on bone mechanical properties. The skeletal response of cortical bone to parathyroid hormone (PTH) is complex and species dependent. Intermittent administration of PTH to rats increases periosteal bone formation but has no known effects on intracortical bone turnover. The effects of continuous PTH on cortical bone are not clearly established. Eighty-eight 6-month-old female Sprague-Dawley rats were divided into 3 control, 6 PTH and 2 PTH withdrawal (WD) groups. They were subcutaneously implanted with osmotic pumps loaded with vehicle or 40 µg/kg BW/d human PTH (1-34) for 1, 3, 5, 7, 14 and 28d. After 7d, PTH was withdrawn from 2 groups of animals for 7 (7d-PTH/7d-WD) and 21d (7d-PTH/21d-WD). Histomorphometry was performed on periosteal and endocortical surfaces of the tibial diaphysis in all groups. Micro-CT of tibiae and mechanical testing by torsion of femora were performed on 28d-PTH and 7d-PTH/21d-WD animals. Continuous PTH increased periosteal and endocortical bone formation, endocortical osteoclast perimeter, and cortical porosity in a time-dependent manner, but did not change the mechanical properties of the femur, possibly due to addition of new bone onto periosteal and endocortical surfaces. Additionally, withdrawal of PTH restored normal cortical porosity and increased maximum torque to failure and stiffness. We conclude that continuous administration of PTH increased cortical porosity in rats without having a detrimental effect on bone mechanical properties.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Articles in Peer-reviewed Journals	Hefferan TE, Evans GL, Lotinun S, Zhang M, Morey-Holton E, Turner RT. "Effect of gender on bone turnover in adult rats during simulated weightlessness." J Appl Physiol (1985). 2003 Nov;95(5):1775-80. https://doi.org/10.1152/japplphysiol.00455.2002 ; PMID: 12882994 , Nov-2003

Articles in Peer-reviewed Journals	Maran A, Hefferan TE, Zhang M, Turner RT. "Unanticipated changes in steady-state mRNA levels for glyceraldehydes-3-phosphate dehydrogenase in rat tibiae." Calcif Tissue Int. 2004 Feb;74(2):204-7. PMID: 14583840 , Feb-2004
Articles in Peer-reviewed Journals	Lotinun S, Evans GL, Bronk JT, Bolander ME, Wronski TJ, Ritman EL, Turner RT. "Continuous parathyroid hormone induces cortical porosity in the rat: Effects on bone turnover and mechanical properties." J Bone Miner Res. 2004 Jul;19(7):1165-71. PMID: 15177000 , Jul-2004
Articles in Peer-reviewed Journals	Sibonga JD, Lotinun S, Evans GL, Pribluda VS, Green SJ, Turner RT. "Dose-response effects of 2-methoxyestradiol on estrogen target tissues in the ovariectomized rat." Endocrinology. 2003 Mar;144(3):785-92. https://doi.org/10.1210/en.2002-220632 ; PMID: 12586754 , Mar-2003